

How well does a current LLM understand Indirect Requests? | Min Htut Win

1 Introduction

People often say things indirectly. For example, someone saying “It’s cold in here” may just be making a comment. However, they may also want someone else to turn on the heater. To understand what the speaker means, it can help a lot to understand what the speaker knows about the situation.

Trott and Bergen (2019) studied this using short stories. In Experiment 1, the speaker either knew about an obstacle or did not know about it. After each story, participants chose between two possible interpretations of the speaker’s final sentence: a request or a literal interpretation. The participants chose the request meaning more often when the speaker was unaware of the obstacle (76%) than when the speaker was aware (30%). Jones et al. (2024) later followed up with a similar task with GPT-3. In their study, GPT-3 did not show the same difference between the Aware and Unaware conditions.

In this project, I test the task again with a current freely available LLM, GPT-5.6 Luna. I also compare the model with reasoning OFF and reasoning ON. This allows me to see whether reasoning changes the way the model uses information about what the speaker knows.

The main research questions are:

1. Does GPT-5.6 Luna use speaker awareness when interpreting indirect requests?
2. Does enabling reasoning change the speaker-awareness effect?

I expected the model to give more request interpretations when the speaker was unaware of the obstacle (H1). I also expected this difference to be larger when reasoning was enabled (H2).

2 Method

I used the first 8 story pairs from Trott and Bergen (2019)’s Experiment 1. This gave 16 stories in total. Each pair had two versions: one where the speaker was aware of the obstacle and one where the speaker was unaware. The final sentence was the same in both versions. The main difference was whether the speaker knew about the obstacle. After each story, GPT-5.6 Luna had to choose between two interpretations of the final sentence. One interpretation was an indirect request and the other was a literal/non-request meaning.

I tested two conditions: reasoning OFF and reasoning ON. For the reasoning-on condition, I used the default medium reasoning level. Each condition was run three times, giving 96 responses in total. I started every new run in a new conversation. This was done so that the model could not use answers from an earlier run. I ran the experiments through OpenRouter using the OpenAI API.

3 Results

The results were very different between reasoning OFF and ON.

With reasoning OFF, the model gave the request interpretation in 45.8% of the Aware cases and 50.0% of the Unaware cases. The difference was only 4.2 percentage points. Fisher's exact test was not significant ($p = 1.000$).

With reasoning ON, the model gave the request interpretation in 8.3% of the Aware cases and 70.8% of the Unaware cases. The difference was 62.5 percentage points. Fisher's exact test was significant ($p < 0.001$).

There was also a significant interaction between reasoning and speaker awareness ($p = 0.003$). This means that the effect of speaker awareness was different when reasoning was OFF and when it was ON.

4 Error Analysis

I also looked at three separate cases where the model gave the wrong answer. In one case, the model fixated on a character's allergy, prioritizing obvious health/safety concerns over the speaker's awareness. In the 2 other cases, it focused more on background knowledge or contextual expectations of the speaker, rather than the speaker's knowledge.

5 Discussion

The biggest difference in the results was between the two reasoning conditions. Without reasoning, GPT-5.6 Luna gave almost the same number of request interpretations in the Aware and Unaware conditions, meaning whether the speaker was aware or not about the situation did not matter to the LLM at all. With reasoning, the difference became much larger.

However, there are also some limitations. First, I only used one LLM. Secondly, I only used 8 story pairs. Lastly, the same stories were used in several runs, so the 96 responses were not 96 completely independent observations.

6 Conclusion

GPT-5.6 Luna showed almost no speaker-awareness effect when reasoning was OFF. When reasoning was ON, a strong effect appeared, with many more request interpretations in the Unaware condition.

The reasoning-on results were found to be similar to the human results from Trott and Bergen (2019). This suggests that enabling reasoning can change how GPT-5.6 Luna interprets indirect requests greatly, to the point that it performs similarly to humans.

GitHub Link: <https://github.com/htutwin/Indirect-Requests-with-LLMs>

AI Use Declaration: AI tools were used for proofreading and improving clarity only.

References

Trott and Bergen (2019): Individual Differences in Mentalizing Capacity Predict Indirect Request Comprehension
Trott and Bergen (2020): When do comprehenders mentalize for pragmatic inference?
Jones et al. (2024): Comparing Humans and Large Language Models on an Experimental Protocol Inventory for Theory of Mind Evaluation (EPITOME)